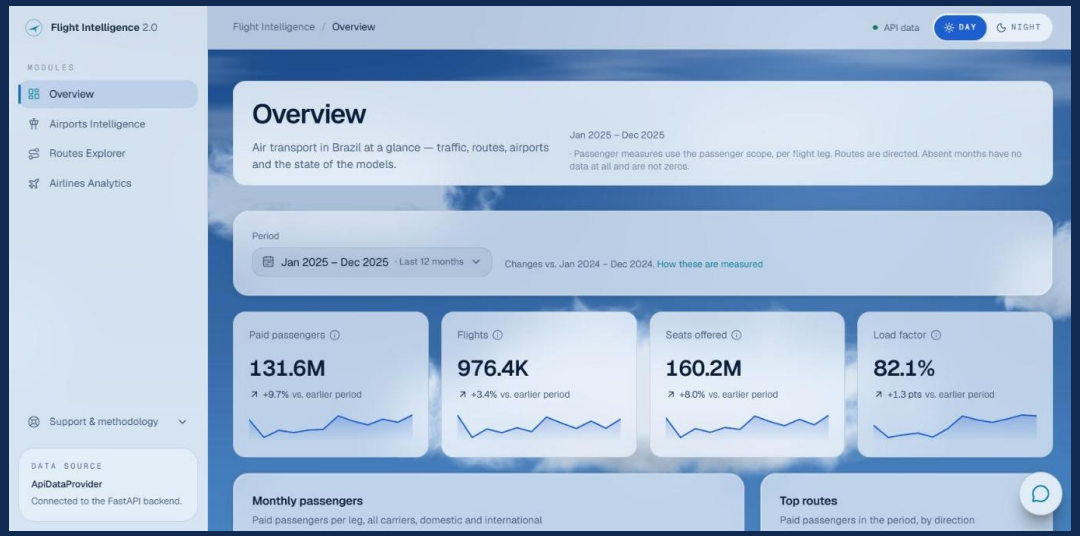
ANAC Flight Intelligence

22,120,241 records.
Traceable analysis.
Honest model evaluation.

Audited ingestion, an analytical dashboard and a tool-grounded AI analyst.



anac-flight-intelligence-api-main.vercel.app

Correct numbers need explicit rules

Different counts

Flights, passenger flights and airport movements are not interchangeable.

Directed routes

CGH → SDU and SDU → CGH are separate markets. Passengers count per leg.

Different ratios

Seat Occupancy = paid passengers / seats. Passenger Load Factor = RPK / ASK.

Unknown is not zero

Four globally absent months remain gaps. Partial annual totals need explicit labels.

Sources: docs/ingestion.md · docs/analytics.md

An audited data foundation

22,120,241

records processed

308

observed months / 312 calendar months

7

analytics tables: 3 dimensions + 4 facts

Bounded chunks → validation flags → atomic Parquet publication → independent DuckDB read-back.

Historical full-run evidence, not a new 20 GB rebuild for this presentation.

Sources: reports/ingestion/quality_report.json · reports/analytics/validation.json

One source, separate analytical paths

ANAC CSV
~20 GB

Parquet
Audited flights

Analytics
Star schema

FastAPI
Docker / Render

Next.js
Vercel

ML

Point-in-time features → LightGBM / baselines → saved 2023 validation → static ML Studio.

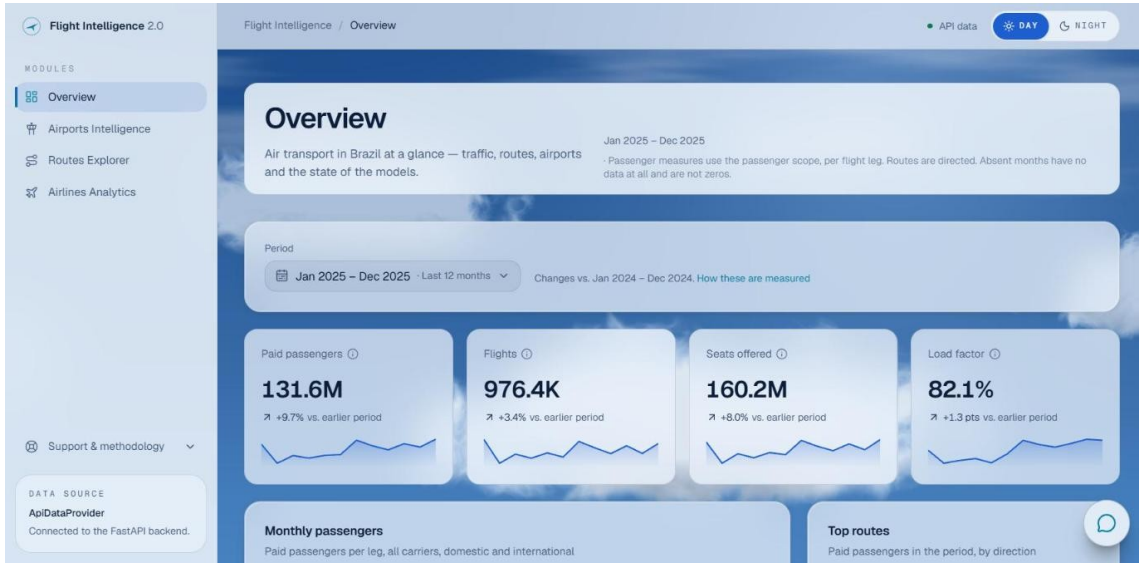
GenAI

Five fixed tools → Groq → heuristic numeric verification → sourced answer. No unrestricted SQL.

The API does not serve operational future predictions.

Sources: docs/ml/en/ML_ARCHITECTURE.md · src/anac/ · web/

A dashboard built around definitions





Market, airports, directed routes and airlines. Period filters, API/DEMO labels, loading/error states and missing-month warnings.

Temporal evaluation, not future forecasts

FIT / EARLY STOP

Purged 2013–2021 fitting; 2022 early stopping. Two-month publication lag.

VALIDATE / FREEZE

Saved selection predictions evaluated on 2023. Final refit boosters include eligible 2023 history.

2024–2025 holdout: partially exposed through earlier annual-total inspection; NOT scored. It is not fully blind.

12 target-month clusters · 2,000 bootstrap draws · intervals describe metric uncertainty.

Sources: configs/models.toml · src/anac/models/train.py · audit.json

Four regressors, ten baseline references

Experiment	LightGBM	Best baseline	Historical verdict
A · Airport movements	10.10%	10.30%	Inconclusive; baseline kept
B · Seat occupancy	6.06 pp	6.54 pp	Pooled CI favors model
D · Passengers	22.24%	25.31%	Historical result; see caveat
D · Seats	22.40%	25.73%	Historical result; see caveat

A: difference 95% CI includes zero. WAPE ≠ accuracy.
B: actual-seat-weighted MAE, not RPK/ASK error. C is NOT trained.

Source: reports/models/module_{a,b,d}.json · errors rounded to two decimals; pp = percentage points.

The route-model caveat stays visible

Historical D training filled globally absent target months with zero. The loader is corrected; frozen v1 models have NOT been retrained.

Impact not measured

11,689 passenger and 11,692 seat fitting rows were affected. No causal performance improvement can be claimed.

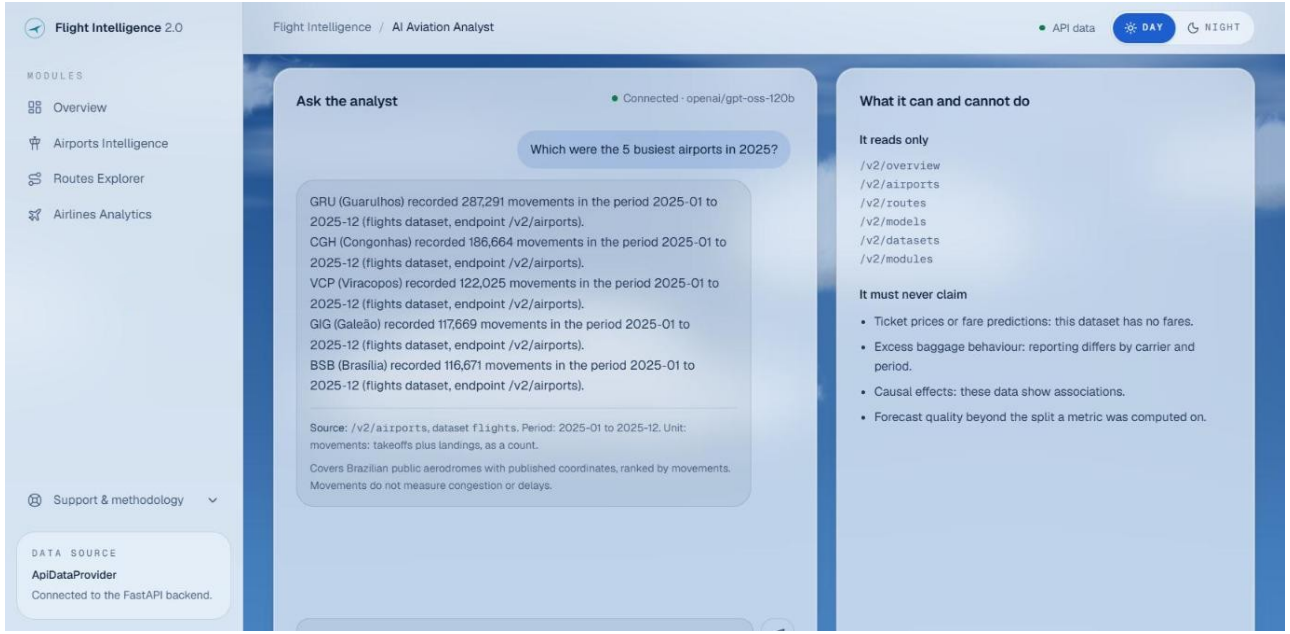
Horizon and bias limits

About −12% aggregate bias. Six-month point errors are worse; paired intervals include zero.

Corrected AP handles ties. Occupancy bias now uses seat weights and explicit units.

Sources: audit.json · docs/ml/en/EXPERIMENT_RESULTS.md. D historical scores are not corrected-model evidence.

Generative AI with controlled data access



Five fixed read-only tools.
Sources attached by code.
Numeric membership checks; not guaranteed semantic attribution.
Quotas: 10/client/hour, 200/day by default, per-process memory.

Server-side key · one corrective attempt before refusal · no fare data.

Source: historical live screenshot; src/anac/agent/ and src/anac/api/ratelimit.py. No new Groq call for this deck.

Verifiable delivery, ready to discuss

368 Python passed · 11 skipped
91 frontend passed
601 ML reconciliation checks

Lint, types and production build passed. Skips require absent local data.
Directed by Caio Bandeira; AI-assisted implementation with Claude Code and Codex. No measured commercial-impact claim.

Live: anac-flight-intelligence-api-main.vercel.app
GitHub: github.com/CaioBandeira7/anac-flight-intelligence-api-main

Sources: docs/portfolio/VALIDATION.md · docs/ml/INTEGRATION.md